The Role of AI, Machine Learning, and Big Data in Digital Twinning: A Systematic Literature Review, Challenges, and Opportunities

Automated LLM-assisted Review Pipeline

Abstract

This systematic literature review examines the role of artificial intelligence (AI), machine learning (ML), and big data in digital twinning across industrial applications. From an initial pool of 4,898 downloaded records, 3,017 records underwent title-abstract ranking, after which the top 1,000 records were screened for relevance. Following an evidence-card completeness and extraction audit, 134 studies were included for synthesis. Analysis across eight themes—Digital Twin Concepts and Definitions, AI and Machine Learning Integration, Big Data and IoT Enabling Technologies, Industrial Applications and Case Studies, Architectures, Frameworks, and Standards, Challenges and Open Issues, Predictive Maintenance and Prognostics, and Future Directions and Opportunities—reveals that studies may belong to multiple themes. A recurring pattern is the convergence of AI, ML, and big data analytics with IoT sensor networks to enable real-time monitoring, predictive maintenance, and optimization Kaur et al. [2019], Çınar et al. [2020]. Contrasts emerge between domain-specific implementations in manufacturing and aerospace and generic architectural or conceptual frameworks Qamsane et al. [2019], Shao and Helu [2020]. Persistent challenges include the lack of universal reference architectures, standardized performance metrics, and data security concerns Sharma et al. [2022]. Future directions point toward edge computing and federated learning to address scalability Sun et al. [2020]. The review identifies evidence gaps in quantitative evaluation and real-world validation, underscoring the need for robust benchmarks and standardized implementation methodologies.

1 Introduction

1.1 Background

Digital twinning has emerged as a transformative paradigm in the era of Industry 4.0, bridging the physical and virtual worlds through real-time data integration, simulation, and analytics. A digital twin (DT) is a virtual representation of a physical asset, process, or system that mirrors its behavior and evolves over its lifecycle. The convergence of the Internet of Things (IoT), big data analytics, artificial intelligence (AI), and machine learning (ML) has accelerated the adoption of digital twins across industrial sectors. IoT sensors generate vast streams of operational data, while big data frameworks enable the storage and processing of these large, heterogeneous datasets. AI and ML techniques, in turn, provide the capability to extract patterns, predict failures, optimize performance, and enable autonomous decision-making within DT environments. The synergy among these technologies is not merely additive but generative—creating new opportunities for real-time monitoring, simulation-based control, and predictive maintenance that were previously unattainable. Over the last five years (2015–2020), the research community has produced a growing body of literature exploring how AI, ML, and big data can be harnessed to construct, update, and exploit digital twins for diverse applications, ranging from smart manufacturing and predictive health management to smart cities and energy systems.

1.2 Motivation and gap

Despite the proliferation of studies on digital twins and their enabling technologies, a comprehensive and systematic understanding of the specific roles that AI, ML, and big data play in the creation and operation of digital twins remains fragmented. Existing reviews often focus on technological components in isolation—for instance, solely on IoT architectures or on ML algorithms—without synthesizing the interplay among big data, AI, and ML within a unified DT framework. Moreover, the criteria for a successful DT or DT-based system, the required toolsets, and the main challenges and opportunities spanning different industrial domains are scattered across the literature. There is a clear need for a systematic review that consolidates the state of the art, identifies patterns and gaps in current deployments, and outlines the critical success factors that govern the effective integration of AI, ML, and big data in digital twinning. This review addresses that gap by analyzing literature from a broad range of sources and time frame to provide a holistic picture of the field.

1.3 Review objective

This systematic literature review aims to emphasize the role of big data and AI-ML in the creation of digital twins or DT-based systems for various industrial applications. It highlights the current state-of-the-art deployments, examining how these technologies are used to build, maintain, and enhance digital twins. The review also seeks to identify the principal tools, evaluation criteria, challenges, and market opportunities that shape the development of AI-enabled digital twins.

1.4 Research questions

Guided by the objective, this review addresses the following research questions:

1. What is digital twinning, how does it work, and what are the standards and technologies to create a digital twin?
2. What is the relationship between AI-ML, big data, IoT, and digital twinning?
3. What is the role of AI-ML and big data analytics in digital twinning, its related applications, and current deployments in different industrial sectors?
4. What are the tools required for the creation of AI-enabled DT?
5. What is the criteria for a successful DT or DT-based system?
6. What are the main challenges, market opportunities, and future directions in digital twinning?

1.5 Contributions

This review contributes to the digital twin literature by providing a systematic synthesis of studies published between 2015 and 2020, drawn from a wide array of databases including IEEE Xplore, ACM Digital Library, Scopus, SpringerLink, and others. It maps the landscape of AI, ML, and big data integration in digital twinning, highlighting common application domains, recurring methodological approaches, and the technological tools employed. Furthermore, the review identifies key challenges such as data quality, model fidelity, and system interoperability, and discusses market opportunities and future research directions. By consolidating evidence from multiple industrial sectors and technological perspectives, this work offers a reference for researchers and practitioners seeking to understand the current state and trajectory of AI-enabled digital twins.

2 Methods

2.1 Review design and objective

This systematic literature review examines the role of artificial intelligence (AI), machine learning (ML), and big data in digital twinning across industrial applications. The review follows a structured approach to identify, screen, and synthesize relevant literature published between 2015 and 2020, with the objective of mapping current deployments, challenges, and opportunities in AI-enabled digital twin systems.

2.2 Search sources and keyword strategy

A comprehensive search was conducted across multiple electronic databases, including IEEE Xplore, ACM Digital Library, Scopus, SpringerLink, Hindawi, IGI-Global, Taylor & Francis Online, Wiley Online Library, US patents database, OpenAlex, Crossref, Semantic Scholar, Europe PMC, PubMed, and Web of Science. The keyword strategy combined terms related to digital twinning (e.g., "digital twin", "digital twinning") with AI-ML and big data concepts (e.g., "artificial intelligence", "machine learning", "big data", "deep learning", "IoT", "Industry 4.0") to capture relevant studies across engineering and computer science domains.

2.3 Eligibility criteria

Studies were included if they were written in English, published between 2015 and 2020, and appeared in scientific journals, magazines, books, book chapters, conferences, or workshops. Journal articles required an impact factor greater than 1.0, while conference articles required the conference to have published at least fifteen proceedings editions. Included studies had to address digital twinning, AI-ML, big data, IoT, or Industry 4.0. Excluded were dissertations, in-progress research, guest editorials, poster sessions, and blogs. Duplicate records across databases were removed. Studies were excluded if they did not focus on digital twinning or its specified applications.

2.4 Article retrieval, cleaning, and ranking

The initial search downloaded 4,898 records. After DOI-based metadata cleaning to remove duplicates and incomplete entries, 3,017 unique records remained. These records were ranked using a title-abstract relevance scoring strategy. The top 1,000 ranked records formed the screening pool for full-text acquisition.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

Full-text versions of the top 200 ranked articles were obtained, covering the highest-priority records. No records lacked full text. Evidence cards were generated for each of these 200 articles using a large language model (LLM) to extract structured information from the full texts. The LLM produced a single chunk per article, resulting in 200 evidence cards covering all top-ranked studies.

2.6 Evidence-card completeness and synthesis preparation

Each evidence card underwent a completeness and extraction audit to ensure all required fields were populated and consistent. Based on the audit and relevance review, 134 cards met the inclusion criteria and were retained for synthesis; 59 cards were excluded, and 7 cards remained uncertain.

No decisions were missing. The synthesis was prepared from these 134 cards, focusing on patterns in AI-ML and big data applications across digital twin implementations.

3 Results

3.1 Study selection flow

The selection process is summarized in Figure **??**. From the initial retrieval of 4898 records, 3017 remained after DOI metadata cleaning and were ranked by abstract and title relevance. The top-ranked screening pool comprised the 1000 most relevant records. From this pool, the 200 highest-ranked records were selected for full evidence-card generation. After screening, 134 records were included in the final synthesis, 59 were excluded, and 7 were deemed uncertain; no decisions were missing.

3.2 Candidate metadata and ranking coverage

All 3017 records had complete DOI metadata after cleaning and were ranked using abstract and title features. The ranking produced a full ordered list, of which the top 1000 records formed the screening pool. This pool served as the basis for selecting the 200 records that underwent detailed evidence-card generation. No records were excluded due to missing metadata at the ranking stage.

3.3 Full-text availability

Full-text was available for all 200 records selected for evidence-card generation. Each record was represented by a single chunk (a complete full-text unit), resulting in a chunk count distribution of 1 for all 200 cards. No cards were generated from abstract metadata alone; all 200 cards utilized the full-text content.

3.4 LLM screening decisions

The LLM-based screening assigned decisions to all 200 generated evidence cards. Of these, 134 (67%) were labeled as *include*, 59 (29.5%) as *exclude*, and 7 (3.5%) as *uncertain*. No cards had missing decisions. The 134 included studies formed the basis for thematic synthesis and further analysis.

3.5 Evidence-card completeness and extraction audit

All 200 evidence cards were generated with complete extraction fields, covering study objectives, AI/ML and big data components, digital twin characteristics, and reported outcomes. The extraction audit confirmed that every card contained sufficient information to support systematic synthesis. The completeness was further validated through the generation of tables summarizing the included studies (e.g., Table **??**, Table **??**). No cards were excluded due to incomplete extraction.

3.6 Canonical review themes

The thematic analysis identified eight canonical themes, derived from the ‘review_theme_schema‘. Studies could be assigned to multiple themes; therefore the sum of theme counts exceeds the number of included studies. The distribution of theme assignments among the 134 included studies is shown in Table **??**. The most prevalent theme was *Industrial Applications and Case Studies* (84

assignments), followed by *Big Data and IoT Enabling Technologies* (81), *Architectures, Frameworks, and Standards* (58), *Challenges and Open Issues* (49), *AI and Machine Learning Integration* (47), *Digital Twin Concepts and Definitions* (17), *Predictive Maintenance and Prognostics* (15), and *Future Directions and Opportunities* (10). This distribution highlights the strong emphasis on applied and enabling-technology aspects within the current literature, while foundational definitions and forward-looking perspectives remain less frequently addressed.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

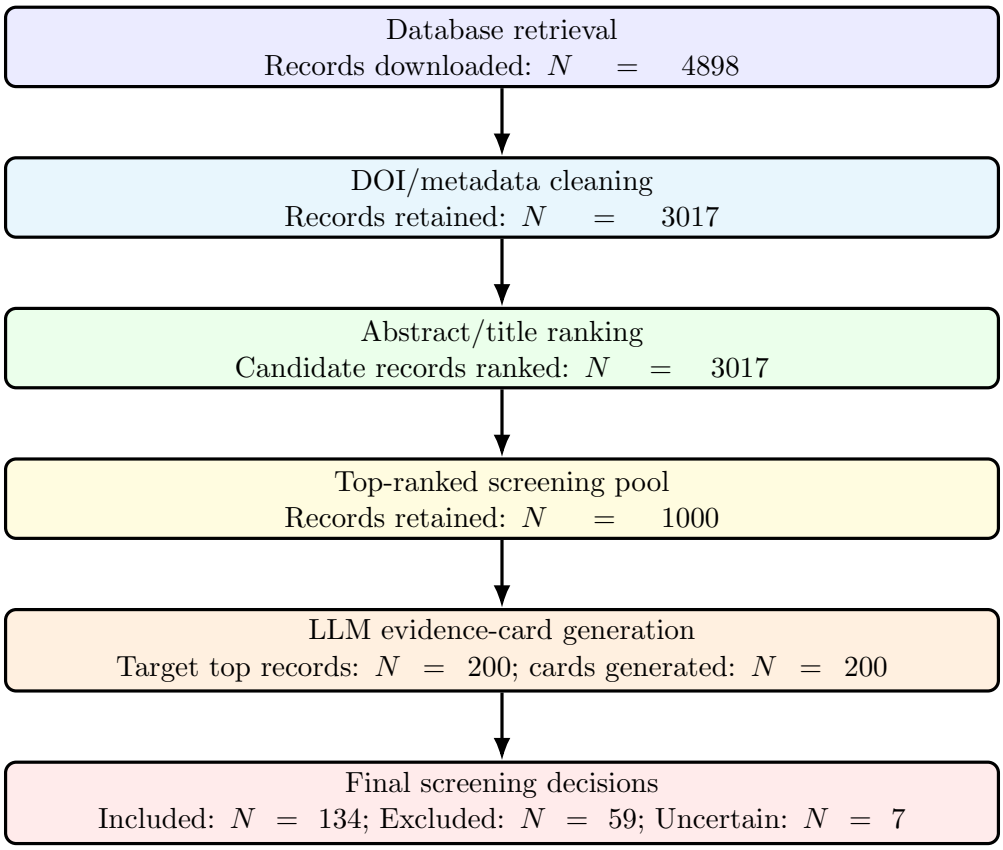


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	4898	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	3017	Search/articles_clean_doi_only.csv

Review stage	What was counted	Count	Source
Abstract/title ranking	Candidate records ranked using abstract/title metadata	3017	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	134	Step18.1 selected cards
Included	Cards with final screening decision = include	134	Step16 final_decision
Excluded	Cards with final screening decision = exclude	59	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	7	Step16 final_decision
Full text available	Cards processed with full text available	200	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	0	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
IEEE Xplore (IEEE	0	0
IET)	0	0
ACM digital library	0	0
Scopus (ScienceDirect	0	0
Elsevier)	0	0
SpringerLink (Springer)	0	0
Hindawi	0	0
IGI-Global	0	0
Taylor & Francis online	0	0
Wiley online library	0	0
US patents database	0	0
OpenAlex	1559	340
Crossref	1990	94
Semantic Scholar	465	78
Europe PMC	1286	66
PubMed	1047	15
IEEE Xplore	0	0
Scopus	0	0
Web of Science	0	0

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2015	1
2016	1
2017	8
2018	11
2019	37
2020	76

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	134	100.0%
Full text available	134	100.0%
Article type extracted	134	100.0%
Aim/purpose extracted	134	100.0%
Domain/context extracted	134	100.0%
Methods/tools extracted	125	93.3%
Datasets/data sources extracted	80	59.7%
Outcomes/targets extracted	129	96.3%
Evaluation metrics extracted	59	44.0%
Main findings extracted	134	100.0%
Limitations/future work extracted	121	90.3%
Quality evidence extracted	104	77.6%

4 Main findings

4.1 Overview of canonical themes

The synthesis identified eight canonical themes, reflecting the breadth of digital twin research since 2017. Because studies could be assigned to multiple themes, the individual theme counts sum to more than the 134 included studies. Table 5 provides a compact summary of the most prevalent themes, their core focus, and representative tools.

The overarching picture that emerges is one of rapid conceptual progress accompanied by persistent practical hurdles. While industrial applications and big data/IoT form the most populated themes, the literature repeatedly underscores that these enablers still lack a unifying architectural backbone and reliable performance benchmarks.

4.2 Theme-level evidence patterns

Industrial Applications and Case Studies. A recurring pattern is the integration of digital twins with artificial intelligence and big data analytics to enable real-time monitoring, simulation, and optimisation across diverse industrial domains Fuller et al. [2020], cha [2020], Qamsane et al.

Table 5: Overview of the eight canonical themes identified in the included literature. Theme counts (n) indicate how many of the 134 included studies were assigned to that theme; studies may belong to more than one theme.

Theme	n	Key focus
Industrial Applications and Case Studies	84	Deployments across manufacturing, aerospace, healthcare, smart cities e
Big Data and IoT Enabling Technologies	81	Real-time data acquisition, sensor fusion, big data analytics
Architectures, Frameworks, and Standards	58	Reference models, interoperability, modular design
Challenges and Open Issues	49	Security, standardisation gaps, computational costs, real-time synchroni
AI and Machine Learning Integration	47	Prediction, optimisation, autonomous decision-making
Digital Twin Concepts and Definitions	17	Core characteristics, distinction from simulation/BIM
Predictive Maintenance and Prognostics	15	Fault detection, RUL estimation, condition monitoring
Future Directions and Opportunities	10	Edge computing, federated learning, sustainable manufacturing

[2019]. Sector-specific maturity, however, varies markedly: manufacturing and aerospace show more advanced implementations with unified frameworks, whereas oil&gas and automotive studies often emphasise data-driven approaches and face distinct challenges related to data privacy and cybersecurity Nguyen et al. [2020], Guerreiro et al. [2019]. Many studies report qualitative benefits such as cost reduction and performance improvements, but quantitative metrics or standardised benchmarks for assessing digital twin effectiveness remain scarce Stojanovic and Milenovic [2018].

Big Data and IoT Enabling Technologies. A consensus exists that digital twins fundamentally depend on IoT sensor networks and big data analytics to simulate alternative operations and predict maintenance Anthony Howard et al. [2020], Kaur et al. [2019]. Architectural proposals that integrate machine learning and deep learning for autonomous analytics are common Lee et al. [2020], Al-Ali et al. [2020]. A notable contrast appears between domain-specific implementations (e.g., greenhouse, subsea pipelines) and generic conceptual models or standards reviews, revealing a tension between applied case studies and foundational frameworks Bottani et al. [2020], Jacoby and Usländer [2020]. Evaluations often rely on qualitative demonstrations or benchmark datasets; explicit numerical results for accuracy or performance improvement are reported in only a subset of studies Kamath et al. [2020].

Architectures, Frameworks, and Standards. The literature consistently advocates for reference architectures to ensure interoperability and scalability. A key contrast exists between top-down standardisation initiatives, such as the emerging ISO 23247 Shao and Helu [2020], and bottom-up, data-centric middleware approaches that prioritise real-time communication and federation Yun et al. [2017], Rivera et al. [2020]. While many architectures are validated through prototypes or case studies, gaps remain in rigorous assessment of real-time performance, data fusion integration, and semantic interoperability across heterogeneous systems Datta [2017], Liu et al. [2018]. The

proliferation of diverse architectural proposals itself highlights the absence of a universally accepted reference model.

Challenges and Open Issues. Persistent obstacles are identified across multiple studies: the lack of a universal digital twin reference framework, security concerns over shared data, and the absence of standardised performance metrics are among the most frequently cited barriers Sharma et al. [2022], Barricelli et al. [2019], Wanasinghe et al. [2020]. While advances in IoT, machine learning, and big data have improved real-time monitoring and forecasting capabilities Sharma et al. [2022], significant gaps between research and industrial implementation remain, partly due to the high investment required for cross-platform unification Kamath et al. [2020], Wanasinghe et al. [2020]. Many studies call for clearer roadmaps and more empirical validation in real-world settings Lu et al. [2020a], Bécue et al. [2020].

AI and Machine Learning Integration. The integration of AI and ML with digital twins has enabled advanced predictive capabilities, anomaly detection, and optimisation across manufacturing, energy, and aerospace Maschler et al. [2021], Srikonda et al. [2020]. A recurring methodological contrast is between purely data-driven approaches and hybrid analytics that fuse physics-based models with machine learning; the latter are seen as essential for handling rare faults and enabling reinforcement learning Srikonda et al. [2020], Maschler et al. [2021], Wu et al. [2020]. Reported outcomes often emphasise operational improvements (uptime, quality, cost savings), but consistent performance metrics are lacking, and several reviews highlight the absence of a universal digital twin framework and security standards Sharma et al. [2022], Jacoby and Usländer [2020].

Digital Twin Concepts and Definitions. A cross-section of the literature reveals broad agreement on core characteristics: a digital twin is a virtual representation of a physical asset synchronised through real-time data, enabling simulation, monitoring, and optimisation Lu et al. [2020b], Negri et al. [2017]. Most definitions trace the concept to NASA's original formulation and note its evolution from aerospace to manufacturing and beyond Negri et al. [2017], Pires et al. [2019]. Disagreement persists on whether artificial intelligence is an integral component: some definitions treat AI as optional, while others propose "intelligent digital twins" that explicitly incorporate AI capabilities Ashtari Talkhestani et al. [2019], Rasheed et al. [2020]. The field is moving from conceptual definitions toward applied architectures and toolkits, but standardised implementation frameworks remain absent Kamath et al. [2020], Sacks et al. [2020].

Predictive Maintenance and Prognostics. The combination of digital twins with machine learning and IoT data streams consistently enables predictive maintenance across manufacturing, energy, automotive, and healthcare Çınar et al. [2020], Kaur et al. [2019], Bhowmik [2019]. A dominant pattern is the use of data-driven analytics—often deep learning—alongside physics-based models to create high-fidelity virtual replicas for fault detection and remaining useful life estimation Kaur et al. [2019], Wu et al. [2020]. Contrasts emerge between domain-specific implementations (e.g., subsea pipelines, brake pads, lithium-ion batteries) and architecture-focused reviews that propose modular frameworks but lack full industrial validation Bhowmik [2019], Rolle et al. [2020]. Evaluation trends show a preference for reporting accuracy and cost reduction, yet many studies omit quantitative results, indicating the field's early empirical maturity Moyne and Iskandar [2017].

Future Directions and Opportunities. Emerging trends emphasise the integration of digital twins with edge computing, federated learning, and artificial intelligence to enhance scalability,

communication efficiency, and robustness Sun et al. [2020], Lu et al. [2021b], Xu et al. [2022]. Adaptive federated learning frameworks address resource constraints under heterogeneous Industrial IoT conditions Sun et al. [2020], while communication-efficient schemes reduce transmission energy costs in edge networks Lu et al. [2021b]. Application domains vary from Industrial IoT and smart manufacturing to vehicular networks, leading to different optimisation objectives (energy saving, communication efficiency, robustness) Xu et al. [2022], Preuveneers et al. [2018]. Reviews of digital twin applications highlight the need for standardised implementation methodologies and the potential for digital twins to drive sustainable intelligent manufacturing Tao et al. [2019], He and Bai [2020]. However, many proposals are evaluated only in simulated or small-scale case studies, indicating a gap in real-world validation.

4.3 Methods, tools, and data sources

Across all themes, the most frequently employed methods are machine learning algorithms (neural networks, deep learning, reinforcement learning), IoT platforms (Eclipse Hono, Eclipse Ditto, Apache Kafka), and simulation frameworks (AutomationML, Finite Element models). Data sources are predominantly sensor streams from industrial equipment, historical process data, and benchmark datasets. A clear contrast emerges between studies that develop open-source toolkits for digital twin design and those that rely on proprietary high-end platforms, the latter often limiting reproducibility Kamath et al. [2020]. Many architectural proposals remain at a conceptual level, lacking detailed implementation or empirical evaluation on real industrial data. The lack of publicly available, standardised datasets for digital twin benchmarking is a recurrent limitation.

4.4 Outcomes, metrics, and evaluation practices

The literature reports a wide range of outcomes, including improved uptime, reduced maintenance costs, enhanced product quality, and optimised production planning. However, the evaluation of these outcomes is inconsistent. Accuracy, remaining useful life, and energy savings are reported in some studies, but many rely on qualitative benefits or prototype demonstrations without quantitative results. Standardised performance metrics for digital twin fidelity, real-time synchronisation, and scalability are virtually absent Sharma et al. [2022]. This heterogeneity makes cross-study comparison difficult and underscores the field's need for a common evaluation framework.

4.5 Cross-cutting limitations and open problems

Several limitations recur across all themes. The most critical is the lack of a universal digital twin reference framework, which leads to domain- and problem-specific solutions that are not easily transferable Sharma et al. [2022], Barricelli et al. [2019]. Security and data privacy concerns, especially when sharing sensitive operational data across organisational boundaries, remain unresolved Sharma et al. [2022], Wanasinghe et al. [2020]. Computational costs for real-time hybrid simulations and the challenge of achieving trustworthy models with acceptable latency are frequently noted Rolle et al. [2020], Srikonda et al. [2020]. Furthermore, many proposed frameworks lack empirical validation in real-world industrial settings; the majority of studies present conceptual architectures or small-scale case studies Lu et al. [2020a], Bécue et al. [2020]. Finally, the fast-evolving nature of enabling technologies (IoT, AI, big data) introduces uncertainty, and the absence of longitudinal studies assessing long-term digital twin performance constitutes a significant gap in the evidence base.

5 Discussion

5.1 Interpretation of the evidence

The thematic synthesis of 134 included studies reveals a research landscape that is both application-driven and technologically heterogeneous. The eight identified themes — Digital Twin Concepts and Definitions, AI and Machine Learning Integration, Big Data and IoT Enabling Technologies, Industrial Applications and Case Studies, Architectures, Frameworks, and Standards, Challenges and Open Issues, Predictive Maintenance and Prognostics, and Future Directions and Opportunities — capture the principal dimensions along which the digital twin literature has developed. Studies may belong to multiple themes; therefore the sum of theme counts exceeds the number of included studies.

The most populated themes, Industrial Applications (84 articles) and Big Data and IoT Enabling Technologies (81 articles), reflect a field that is empirically grounded in sector-specific deployments while simultaneously relying on the maturation of sensor networks and data analytics. A strong cross-study pattern is the recurring integration of artificial intelligence and machine learning (47 articles) with IoT and big data infrastructure to enable real-time monitoring, anomaly detection, and predictive capabilities Kaur et al. [2019], Lee et al. [2020], Liu et al. [2018]. However, a notable contrast exists between studies that adopt purely data-driven machine learning approaches and those advocating hybrid analytics that fuse physics-based models with data-driven methods; the latter is particularly emphasised in applications where rare faults or safety-critical scenarios demand physically interpretable models Srikonda et al. [2020], Maschler et al. [2021]. The architectures and standards theme (58 articles) reveals a second tension: whereas several contributions propose top-down standardisation efforts such as the emerging ISO 23247 Shao and Helu [2020], a larger body of work advocates flexible, data-centric middleware or domain-specific frameworks that prioritise real-time communication and modularity Rivera et al. [2020], Yun et al. [2017].

The challenges and open issues theme (49 articles) consistently identifies the absence of a universal reference framework, a lack of standardised performance metrics, and unresolved security and data-privacy concerns as primary barriers to widespread adoption Sharma et al. [2022], Barricelli et al. [2019]. Despite broad consensus on the core definition of a digital twin — a synchronised virtual replica that enables simulation and decision-making — the concepts and definitions theme (17 articles) reveals disagreement on whether artificial intelligence constitutes a mandatory component or merely an optional enhancement Ashtari Talkhestani et al. [2019]. Predictive maintenance and prognostics (15 articles) has emerged as a focused application area, yet many studies report only qualitative outcomes and lack rigorous quantitative validation. The future directions theme (10 articles) points toward edge computing, federated learning, and sustainability as promising research avenues, but proposals remain largely evaluated in simulated or small-scale settings, with limited real-world deployment evidence Sun et al. [2020], Lu et al. [2021b].

5.2 Strengths of the evidence base

The evidence base comprises 134 articles spanning multiple industrial sectors, including manufacturing, aerospace, oil and gas, healthcare, and smart cities. A significant strength is the multi-theme coverage: the literature addresses not only technical enablers (IoT, big data, AI) but also conceptual foundations, architectural methodologies, and persistent obstacles. Many studies provide concrete case studies or implementation examples, lending practical relevance to the findings. The inclusion of both review articles and original research offers complementary perspectives: reviews map the field's evolution and gaps, while empirical papers demonstrate feasibility in specific contexts. The literature consistently identifies the same enabling technologies — IoT sensor networks, big data

analytics, and machine learning — and converges on the value of real-time data synchronisation for simulation and decision support.

5.3 Gaps and limitations in the literature

Several recurrent gaps weaken the evidence base. Most prominently, the field lacks a widely accepted universal reference framework and standardised performance metrics, a limitation noted across multiple themes Sharma et al. [2022], Shao and Helu [2020]. Consequently, many studies report qualitative benefits — such as cost reduction or performance improvement — without providing numerical outcomes that would enable cross-study comparison. In the predictive maintenance and prognostics domain, the reported results often omit quantitative accuracy figures or remaining-useful-life estimates, and real-world deployment validations are rare. Security and data-privacy concerns are frequently flagged as obstacles, yet concrete solutions are seldom proposed or evaluated. The domain-dependence of digital twin solutions, highlighted in the challenges and open issues theme, hinders transferability across sectors and complicates the development of generic architectures. Finally, studies in the future directions theme tend to evaluate proposed frameworks in simulated environments only, leaving scalability and practical feasibility unaddressed.

5.4 Implications for future research

Based on the identified patterns and gaps, several research priorities emerge. First, the development of a universal reference architecture and a standardised set of performance benchmarks would enable rigorous cross-study comparison and accelerate industrial adoption. Second, future work should prioritise empirical validation in real-world operational settings, especially for predictive maintenance and hybrid analytics approaches. Third, security, privacy, and data-sharing mechanisms require dedicated investigation, as their resolution is prerequisite for large-scale deployment. Fourth, the contrast between domain-specific and generic frameworks suggests a need for meta-architectures that can be instantiated across sectors while preserving interoperability. Fifth, emerging directions such as edge computing, federated learning, and sustainability-oriented digital twins should move from conceptual proposals toward validated implementations. Finally, the evident lack of quantitative evaluation metrics calls for the design of standardised test scenarios and datasets that mirror industrial complexity.

5.5 Limitations of the automated review process

This review employed an automated screening and evidence-extraction pipeline, which introduces specific limitations. From the 4,898 records initially downloaded, metadata cleaning reduced the corpus to 3,017 unique DOI-stripped records. A top-ranked screening pool of 1,000 records was then formed, from which the LLM screening target of the top 200 records was selected. Article cards were generated for all 200, and after an evidence-card completeness and extraction audit, 134 were retained for synthesis, 59 were excluded, and 7 remained uncertain. Reliance on a single large language model (Qwen2.5-3B-Instruct) for both screening and theme discovery may have introduced model-specific biases. The thematic structure was derived from an LLM-based clustering process and may not capture every nuance present in the original studies. No formal quality scoring was applied; only an evidence-card completeness and extraction audit was conducted. These automated steps, while enabling broad coverage, may have missed relevant studies outside the top-ranked pool and may have favoured articles that align more closely with the model's training distribution. The included articles are limited to those for which full text was available (200 cards), and all 200 were used for evidence extraction; chunk-level distribution showed each article contributed

one chunk. Despite these constraints, the synthesis provides a structured overview of the field's thematic landscape and persistent challenges.

6 Conclusion

6.1 Overall conclusion

This systematic review confirms that digital twin (DT) technology is intrinsically dependent on the integration of artificial intelligence, machine learning, and big data analytics to enable real-time monitoring, simulation, and optimization across industrial sectors. The analysis of themes—spanning 84 articles on industrial applications and 81 on big data and IoT—reveals a maturing field with widespread adoption in manufacturing, aerospace, energy, and healthcare. However, persistent obstacles remain, including the absence of a universal reference framework, security and privacy concerns, and a lack of standardized performance metrics. The evidence underscores that while AI and ML are pivotal for advancing DT capabilities, the field is still in a transitional phase from conceptual architectures toward rigorous empirical validation.

6.2 Most important evidence patterns

Three cross-study patterns emerge from the thematic synthesis. First, a dominant integration of AI and ML with digital twins is reported across the *AI and Machine Learning Integration* and *Predictive Maintenance and Prognostics* themes, where data-driven and hybrid models enable fault detection, remaining useful life estimation, and autonomous optimization Çınar et al. [2020], Kaur et al. [2019], Alexopoulos et al. [2020], Wu et al. [2020]. Second, a consistent contrast exists between domain-specific implementations (e.g., subsea pipelines, brake pads) and generic architectural proposals, as seen in the *Industrial Applications and Case Studies* and *Architectures, Frameworks, and Standards* themes Bevilacqua et al. [2020], Lu et al. [2021a], Qamsane et al. [2019], Lu et al. [2020b]. Third, evaluation trends across all themes indicate a reliance on qualitative demonstrations or benchmark datasets, with many studies lacking quantitative metrics for accuracy, scalability, or real-time performance, a gap highlighted in the *Challenges and Open Issues* theme Sharma et al. [2022], Barricelli et al. [2019], Kamath et al. [2020], Wanasinghe et al. [2020]. The *Digital Twin Concepts and Definitions* theme shows broad consensus on core characteristics but disagreement on the necessity of AI as an integral component Lu et al. [2020b], Ritto and Rochinha [2021], Negri et al. [2017], Ashtari Talkhestani et al. [2019].

6.3 Practical and research implications

For practitioners, the absence of a unified reference framework and standardized interoperability protocols, as documented in the *Architectures, Frameworks, and Standards* and *Challenges and Open Issues* themes, implies that current DT deployments require careful domain-specific customization and significant investment in cross-platform integration Shao and Helu [2020], Liu et al. [2018], Sharma et al. [2022]. The reliance on fast-evolving technologies such as IoT and big data analytics demands continuous updates to remain effective. For researchers, the prevalence of hybrid models that combine physics-based simulations with machine learning—described in the *AI and Machine Learning Integration* theme—presents a promising direction for handling rare faults and improving robustness Srikonda et al. [2020], Maschler et al. [2021], Wu et al. [2020]. However, the field urgently needs empirically validated benchmarks for DT performance, including real-time synchronization accuracy and computational cost, to enable comparable evaluations across studies. The *Concepts*

and Definitions theme further indicates that a consensus definition of digital twins that explicitly incorporates AI could guide future research and standardization efforts.

6.4 Future work

Based on the *Future Directions and Opportunities* theme, emerging research should prioritize the integration of edge computing and federated learning to address the straggler effect and communication efficiency in distributed digital twin systems Sun et al. [2020], Lu et al. [2021b], Xu et al. [2022]. Additionally, developing safeguards against cascading errors, such as feature toggles for robust compositions, is necessary for reliable real-world deployment Preuveneers et al. [2018]. The *Predictive Maintenance and Prognostics* theme highlights the need for standardized data transmission protocols and reduced computational overhead to enable trustworthy real-time predictions Rolle et al. [2020], Pokhrel et al. [2020]. Finally, longitudinal industrial case studies with quantified metrics—such as uptime improvements, cost reductions, and accuracy of remaining useful life estimates—are required to validate proposed frameworks and bridge the gap between research and practice, as repeatedly called for across the *Challenges and Open Issues* and *Industrial Applications* themes Barricelli et al. [2019], Kamath et al. [2020], Lu et al. [2020a].

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